Convert NMEA Coordinates into Google Earth Application Note

GNSS Module Series

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Quectel Wireless Solutions Co., Ltd.
Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China
Tel: +86 21 5108 6236
Email: info@quectel.com

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About the Document

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Contents

About the Document 3
Contents 4
1 Introduction 5
2 Convert a NMEA Logfile for Usage in Google Earth 6
3 Real Time Tracking with Google Earth 8
4 Calculating GPS Latitude/Longitude Position from NMEA Strings 9

1 Introduction

This document describes displaying NMEA coordinates in Google Earth. If you do not have Google Earth installed on your Computer, then please download and install it. See *Earth Versions – Google Earth*.

The following subjects are described:

1 Convert a NMEA Logfile for Usage in Google Earth.
2 Real Time Tracking with Google Earth.
3 Calculating GPS Latitude/Longitude Position from NMEA Strings.

2 Convert a NMEA Logfile for Usage in Google Earth

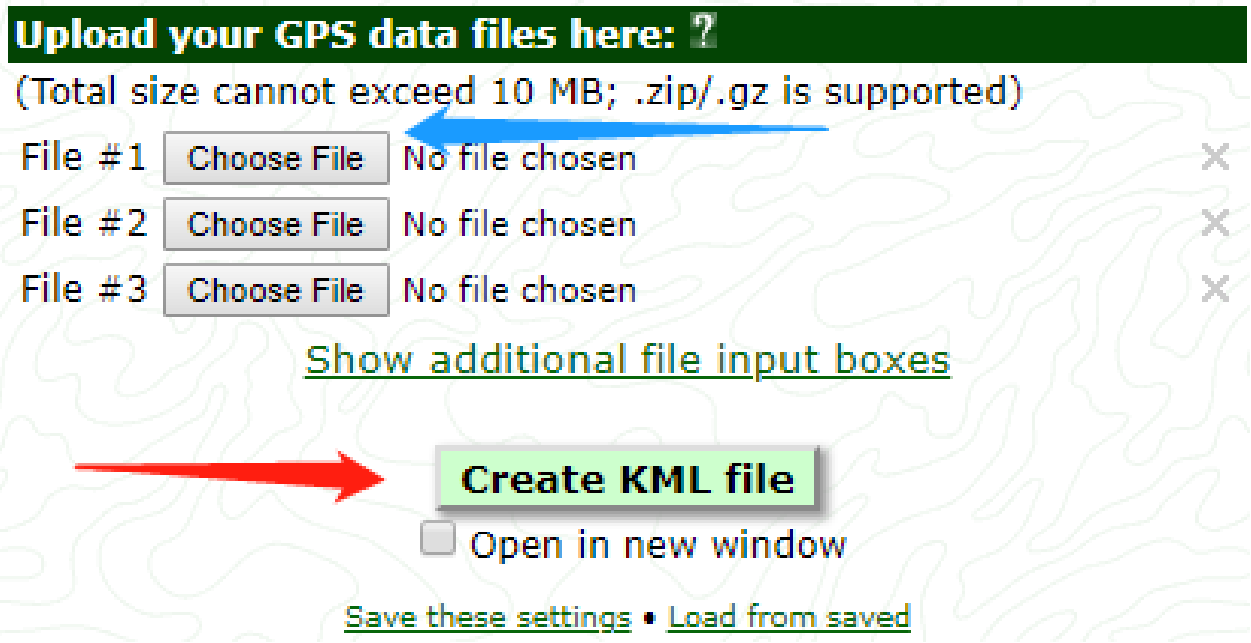
The GNSS module output messages can be saved in txt format, the s/w application Google earth uses KMZ or KML format. The conversion from txt format to KML/KMZ is described below:

The following rules should be followed:

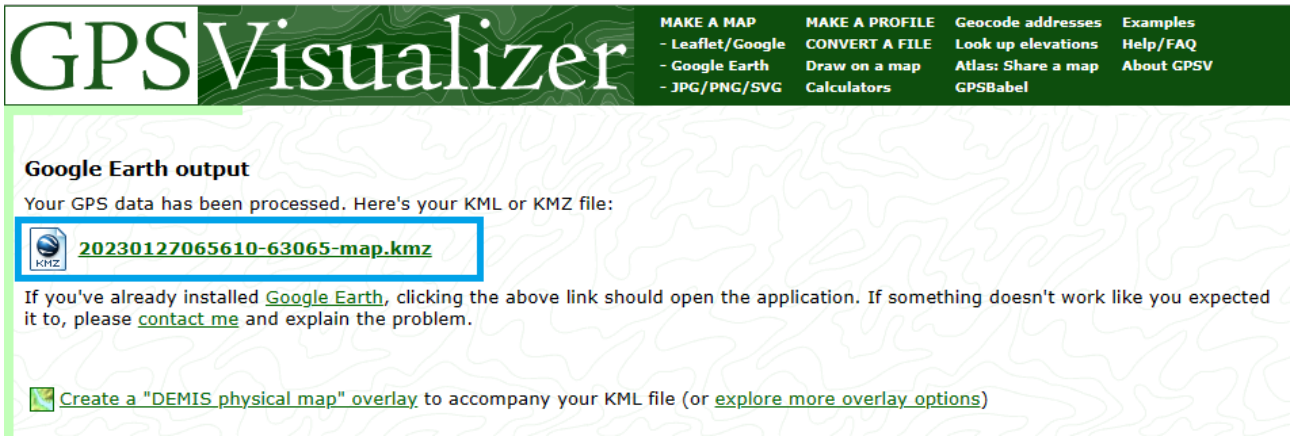
- Majority of mapping applications can only display valid position fix points.
- Once your GNSS/GPS module is in tracking mode (as soon as it has a position fix) please save the NMEA log file into **logfile.txt** format (the logfile can be any name you want but in a txt format). The log must contain only the standard NMEA messages. Any terminal or software utility can be used to save the GNSS module messages. An example of such log is given below:

```
$GNRMC,113609.000,A,3209.359787,N,03455.234623,E,0.00,4.19,211222,,,A,V*00
$GNVTG,4.19,T,,M,0.00,N,0.00,K,A*2F
$GNGGA,113609.000,3209.359787,N,03455.234623,E,1,15,0.71,40.132,M,17.189,M,,*44
$GNGSA,A,3,14,05,19,15,13,17,30,07,20,,,,1.50,0.71,1.32,1*0D
$GNGSA,A,3,66,85,75,76,65,72,,,,,,,1.50,0.71,1.32,2*08
$GPGSV,3,1,11,14,68,029,42,13,56,321,39,30,50,057,44,40,46,145,37,0*6C
$GPGSV,3,2,11,17,40,134,44,05,29,254,37,15,27,315,37,19,20,169,35,0*69
$GPGSV,3,3,11,07,20,082,40,20,17,219,28,09,02,141,34,0*52
$GLGSV,2,1,08,65,67,232,42,75,48,009,46,76,40,282,43,66,33,324,43,1*79
$GLGSV,2,2,08,72,26,172,41,85,13,107,42,74,08,051,29,81,,,,1*42
$GNGLL,3209.359787,N,03455.234623,E,113609.000,A,A*41
$GNRMC,113610.000,A,3209.359787,N,03455.234623,E,0.00,4.19,211222,,,A,V*08
$GNVTG,4.19,T,,M,0.00,N,0.00,K,A*2F
$GNGGA,113610.000,3209.359787,N,03455.234623,E,1,15,0.71,40.132,M,17.189,M,,*4C
$GNGSA,A,3,14,05,19,15,13,17,30,07,20,,,,1.50,0.71,1.32,1*0D
```

- To convert the NMEA log into a KML or KMZ format, please use the following web site: *https://www.gpsvisualizer.com/map_input?form=googleearth*
- On the “GPS Visualizer” site, select “Choose File”.



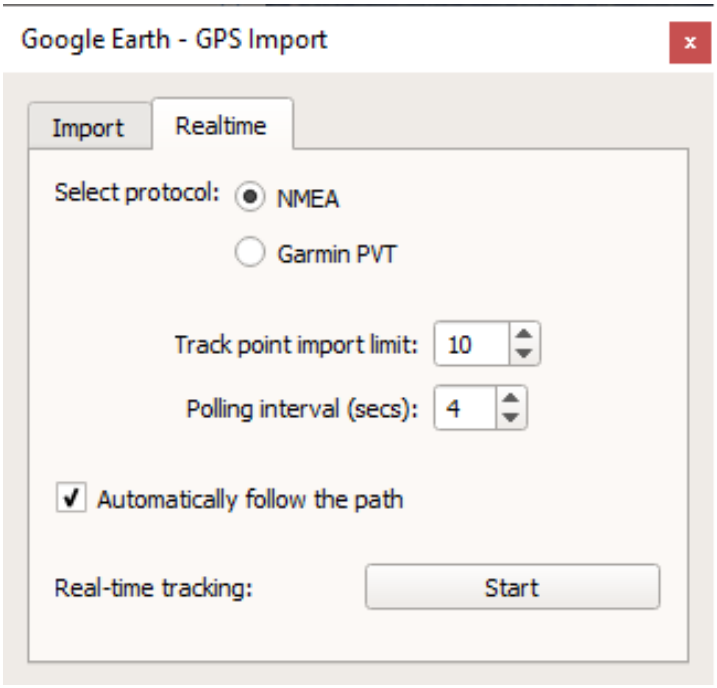
- Select your txt log file that you want to convert and click “Create KML file”. KML is a format supported by Google Earth. KMZ is a compressed format of KML also supported by Google Earth.
- The visualizer s/w utility will generate a file in KML or KMZ format and you will get the following screen:



- Click on the KMZ file link. Your browser will then download the KMZ file to your PC.
- On your PC, once you click on this file it will be opened and displayed over Google Earth.

3 Real Time Tracking with Google Earth

- Run your dedicated module utility software and establish a connection with your GNSS module using the NMEA protocol.
- Once the module is in 2D/3D fix, select the “Disconnect” button in the application. Do not physically disconnect the USB cable from the PC. The module must remain connected to the PC.
- Open Google Earth software utility on your PC.
- From the menu select Tools → GPS. A menu window will appear.
- In the “Realtime” section – please select NMEA.
- Check the “Automatically follow the path” checkbox.
- Click on Start.



Now you can view the tracking path over Google Earth.

4 Calculating GPS Latitude/Longitude Position from NMEA Strings

The coordinates from the module's NMEA output string messages, are different from the Google Earth coordinates format.

As an example, the GNSS modules output following NMEA string:

```
$GNRMC,074804.000,A,4336.7862,N,00657.7702,E,0.00,0.00,130123,,,D,V*0D
$GPGGA,074804.000,4336.7862,N,00657.7702,E,2,17,0.67,126.8,M,48.6,M,,*6F
```

The format of the NMEA latitude and longitude coordinates is in WGS84 Datum: degrees and minutes format (DDMM.mmmm).

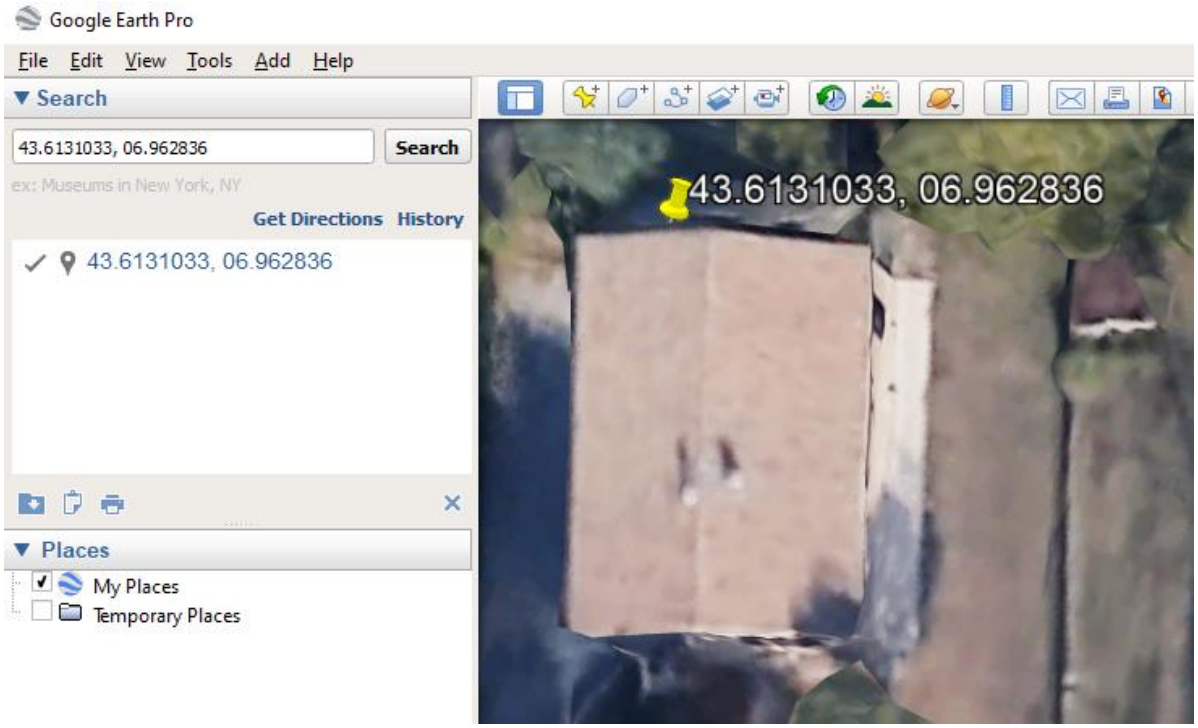
The format of the Google Earth coordinate formats are in Decimal Degrees.

For example, the following NMEA string values for a valid position is:

```
4336.7862 N,00657.7702 E
Converting them into decimal degree values:
```

	NMEA	Calculation Formula	Google Earth
Latitude	4336.7862 N	43 degrees + 36.7862/60 minutes	43.6131033 N
Longitude	00657.7702 E	06 degrees + 57.7702/60 minutes	06.962836 E

These coordinates 43.6131033, 06.962836 can be used in Google Earth and provide you with the correct location. See the image below.


Google Earth Pro
File Edit View Tools Add Help
Search
43.6131033, 06.962836
Search
ex: Museums in New York, NY
Get Directions History
43.6131033, 06.962836
Places
My Places
Temporary Places
43.6131033, 06.962836